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π-Hole Interactions Involving Nitro Compounds: Directionality of Nitrate Esters

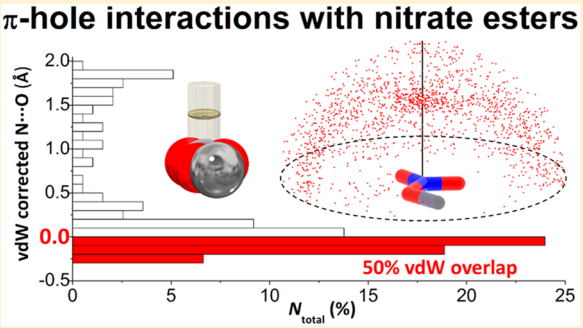
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Supporting Information

ABSTRACT: The MEPs of a variety of nitro compounds (R−NO_2) suggest the existence of a π-hole with a potential of up to +54 kcal/mol in **10** (R = CF_3). Several of these nitro compounds were considered as partners for anions (F^-, Cl^-, NC^-) and the electron rich molecules acetonitrile and dimethyl ether. In most cases a π-hole complex was obtained with calculated binding energies of up to 20 kcal/mol with anions and 5 kcal/mol with the neural molecules. A thorough analysis of the CSD revealed that nitrate esters are highly directional π-holes in the solid state, for at least sp^2 O atoms. This was further illustrated by highlighting several crystal structures where more than 0.2 Å van der Waals overlap was observed between the N atom of the nitrate ester and an electron rich atom like oxygen.



INTRODUCTION

Underpinning molecular recognition is a concert of intermolecular interactions such as hydrogen and halogen bonding interactions.[1−4] The common physical basis of these forces is that electron rich entities can interact with a so-called "σ-hole", which is a positive electrostatic potential along the vector of a covalent bond (e.g., C−H or X−I).[5−7] Positive electrostatic potentials can also be found on electron deficient π-systems and these "π-holes"[8−11] can also attract electron rich species in a directional fashion in the solid state.[8,12,13] We have recently shown that this rationale applies to π-holes in very common nitro-compounds such as nitromethane and nitroaromatics.[8] This led us to question what other nitro-bearing molecules might display such properties[14−18] and to embark on the combined theoretical and Cambridge Structure Database (CSD) analyses reported herein.

RESULTS AND DISCUSSION

We considered various entities for R in R−NO_2, including some that have previously been subjected to theoretical inquiries.[19−22] Shown in Figure 1 is an overview of the molecular electrostatic potential maps of the nitro-compounds **1**−**12**[23] was used to construct the atomic coordinates of starting geometries and to perform the calculations reported herein). While some of these structures are more common than others, all have been reported in the literature[24]. In all except **1** (R = H) there seems to be a π-hole and the potential varies between about +27 kcal/mol in **9** (R = OCH_3) to +54 kcal/mol in **10** (R = CF_3). The π-hole in the halogen compounds **2**−**4** is larger than the σ-hole at the end of the N−F/Cl/Br vector (not shown) and both are about the same in **5** (R = I). It is interesting to note that in structures **10** and **11** the π-hole seems to merge with the σ-hole of adjacent C−H/F, resulting in a positive potential that peaks in the middle of the N−C bond. The π-hole is also slightly translated in the conjugated systems **6**, **7**, and **12**.

To probe the possible association of some of these π-hole bearing molecules with electron rich partners, we considered F^-, Cl^-, NC^-, acetonitrile, and dimethyl ether as "guest" entities. Table 1 contains an overview of the results and Figure 2 contains graphical renderings of some representative examples. Unexpectedly, **2**+F^- converged to the unknown anion $[O_2NF_2]^-$ (not present in the Reaxys database or in the CSD) in which the Mulliken charges (RI-MP2/cc-pVTZ) are +0.42 (N), − 0.25 (O), and −0.46 (F).[25]

Also unforeseen was the location of the anions in **2**+Cl^- and **2**+$^-$CN, which nested themselves in between the nitro's O atoms but away from the nitrate's N atom (F−N···Cl^- angle = 142° in **2**+Cl^-; see Figure 2a). The N−F distance was concurrently elongated from 1.536 Å in FNO_2 to about 1.71 Å in the two complexes. The "Atoms in Molecules"[26] analysis of [**2** + Cl^-]$^-$ clearly reveals bond paths between Cl^- and the two O atoms in **2** (Figure S1) as a consequence of the geometric arrangement of the complex (shorter O···Cl than N···Cl distances).

With **6** (R = CN) the anions (X) attack the nitrile's C atom to form complexes of the type $[O_2NC(N)-X]^-$; an example of such a geometry is shown in Figure 2b for **6**+$^-$CN. Unsurprisingly, the F^- anion attacks **8** to form **2**, whereby trifluoromethoxide acts as a leaving group which then forms a

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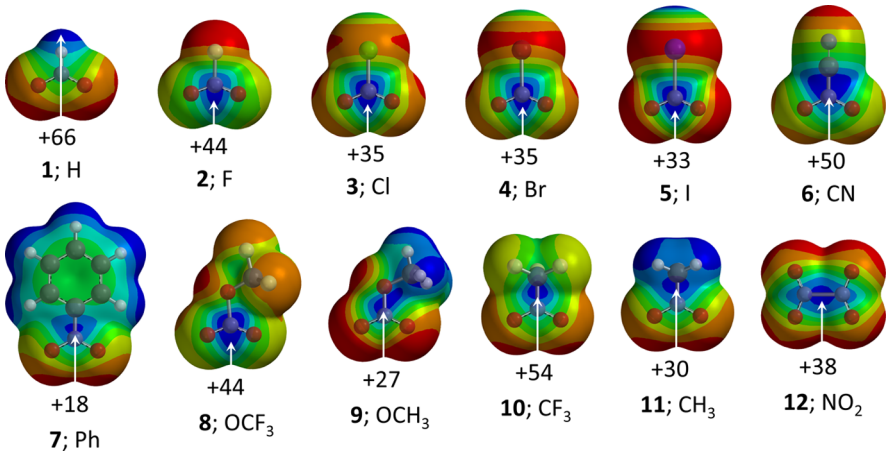


Figure 1. Molecular electrostatic potential maps (MEPs; blue = positive, red = negative) of R-NO_2 molecules derived from MP2/6-311+G** energy optimized geometries. The energetic values are in kcal/mol.

Table 1. Interaction Energies (without and with Basis Set Superposition Error Correction for the Noncovalent Complexes) and Equilibrium Distances at the MP2/6-311+G Level of Theory of Complexes between 2, 6, 8, 10, and 12 and Electron Rich Partners**[a]

π-hole "host"	electron rich "guest" (X)	ΔE (kcal/mol)	ΔE_{BSSE} (kcal/mol)	N···X (Å)	structure[b]
2[20−22] (R = F)	F^-	−22.7	–	1.783	$[O_2NF_2]^-$
	Cl^-	−11.6	−8.9	3.205	$[\mathbf{2}^{2O}\cdots Cl^-]^-$
	NC^-	−11.1	−9.6	2.979	$[\mathbf{2}^{2O}\cdots{}^-CN]^-$
	CH_3CN	−6.02	−3.8	2.932	$[\mathbf{2}^{N}\cdots NCCH_3]$
	$O(CH_3)_2$	−7.05	−3.9	2.642	$[\mathbf{2}^{N}\cdots O(CH_3)_2]$
6[22] (R = CN)	F^-	−43.3	–	1.490[c]	$[O_2NC(N)-F]^-$
	Cl^-	−23.0	–	2.306[c]	$[O_2NC(N)-Cl]^-$
	NC^-	−50.9	–	1.386[c]	$[O_2NC(N)-CN]^-$
	CH_3CN	−6.31	−4.7	2.859	$[\mathbf{6}^{N}\cdots NCCH_3]$
	$O(CH_3)_2$	−8.79	−5.3	2.674	$[\mathbf{6}^{N}\cdots O(CH_3)_2]$
8 (R = OCF_3)	F^-	−53.3[c]	–[d]	1.693[d]	$[\mathbf{2}^{N}\cdots{}^-OCF_3]^-$
	Cl^-	−25.5[d]	–[e]	2.884[e]	$[\mathbf{8}^{2O}-Cl\cdots{}^-OCF_3]^-$
	NC^-	−27.4[e]	–[f]	2.573[f]	$[\mathbf{8}^{2O}-CN\cdots{}^-OCF_3]^-$
	CH_3CN	−7.51	−4.1	2.877	$[\mathbf{8}^{N}\cdots NCCH_3]$
	$O(CH_3)_2$	−8.48	−4.5	2.627	$[\mathbf{8}^{N}\cdots O(CH_3)_2]$
10[19] (R = CF_3)	F^-	−23.5	−18.4	2.197	$[\mathbf{10}^{N}\cdots F^-]^-$
	Cl^-	−14.1	−8.3	3.030	$[\mathbf{10}^{N}\cdots Cl^-]^-$
	NC^-	−11.5	−9.4	2.940	$[\mathbf{10}^{N}\cdots{}^-CN]^-$
	CH_3CN	−6.53	−3.4	2.979	$[\mathbf{10}^{N}\cdots NCCH_3]$
	$O(CH_3)_2$	−8.81	−4.5	2.675	$[\mathbf{10}^{N}\cdots O(CH_3)_2]$
12[22] (R = NO_2)	F^-	−25.0	−20.0	2.248	$[\mathbf{12}^{N}\cdots F^-]^-$
	Cl^-	−15.9	−10.3	3.018	$[\mathbf{12}^{N}\cdots Cl^-]^-$
	NC^-	−13.0	−10.8	2.906	$[\mathbf{12}^{N}\cdots{}^-CN]^-$
	CH_3CN	−6.38	−4.1	2.893	$[\mathbf{12}^{N}\cdots NCCH_3]$
	$O(CH_3)_2$	−9.58	−5.0	2.642	$[\mathbf{12}^{N}\cdots O(CH_3)_2]$

[a]In the starting geometry the partner was placed directly above the R−NO_2 N-atom with an R−N···X angle of 90°. Energies are in kcal/mol. [b]These notations signify the formation of a new molecule or specify, in superscript, if the "guest" interacts with the N or (is bound to) the two O atoms of the nitro compound ("−" indicates chemical bonding and "···" a noncovalent interaction). [c]Covalent bond with C. [d]Binding energy (BSSE corrected) of the noncovalent complex is −9.3 kcal/mol with a N···$^-OCF_3$ distance of 2.834 Å. [e]Binding energy (BSSE corrected) of the noncovalent complex is −17.5 kcal/mol with a N···$^-OCF_3$ distance of 2.238 Å. [f]Binding energy (BSSE corrected) of the noncovalent complex is −17.2 kcal/mol with a N···$^-OCF_3$ distance of 2.283 Å.

π-hole complex with **2** (−9.3 kcal/mol). With **8**+Cl^-/NC^-, the anions drifted away from the N atom to become bifurcated in between the nitrate O's (see Figure 2c for **8**+Cl^-). These unprecedented four-membered ring systems are structural isomers of **3** and **6**. Computations of isomers of **2**−**6** suggests these rings might exist, and also have a π-hole on N (about +45 kcal/mol), but are less stable than their noncyclic isomers **2**−**6** by about 2−8 kcal/mol (see Figure S2). These isomers shed some light on the complexes obtained when computing **2**+Cl^-/NC^- (vide infra); it seems likely that the ring formation is hampered because F^- is a poor leaving group.

When the anionic donors are combined with compounds **10** and **12**, the anticipated anionic π-hole complexes (see Figure 2d, for example, $[\mathbf{10}\cdots Cl^-]^-$) are obtained and the favorable binding energies are in the order of strong hydrogen bonding (15−40 kcal/mol).[27] Likewise, with the neutral electron donors

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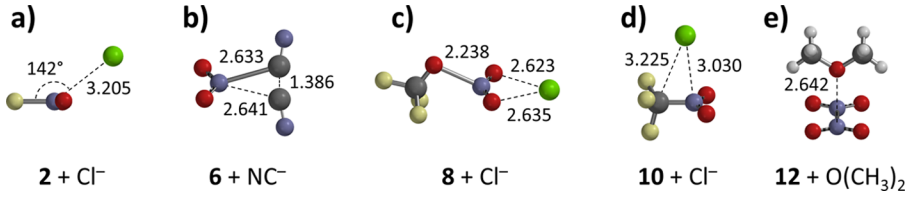


Figure 2. Perspective views of selected complexes (see also Table 1). Distances are in Å.

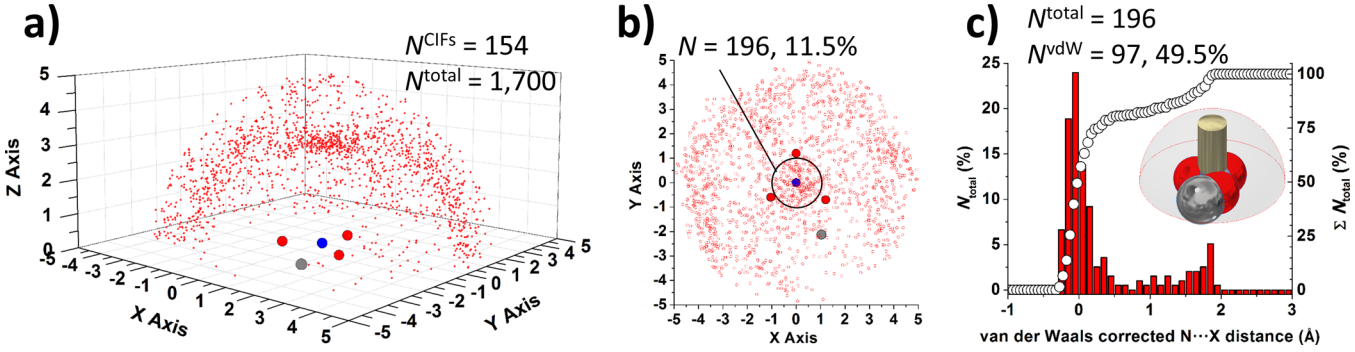


Figure 3. Perspective (a) and top (b) views of the distribution of sp^2 O atoms enveloping a central nitrate ester as found within the CSD with N···O ≤ 5 Å. The data characterized by a parallel displacement of ≤1 Å from N (black circle in b and golden body in the inset in c) were plotted as a function of the van der Waals corrected N···O distance (c).

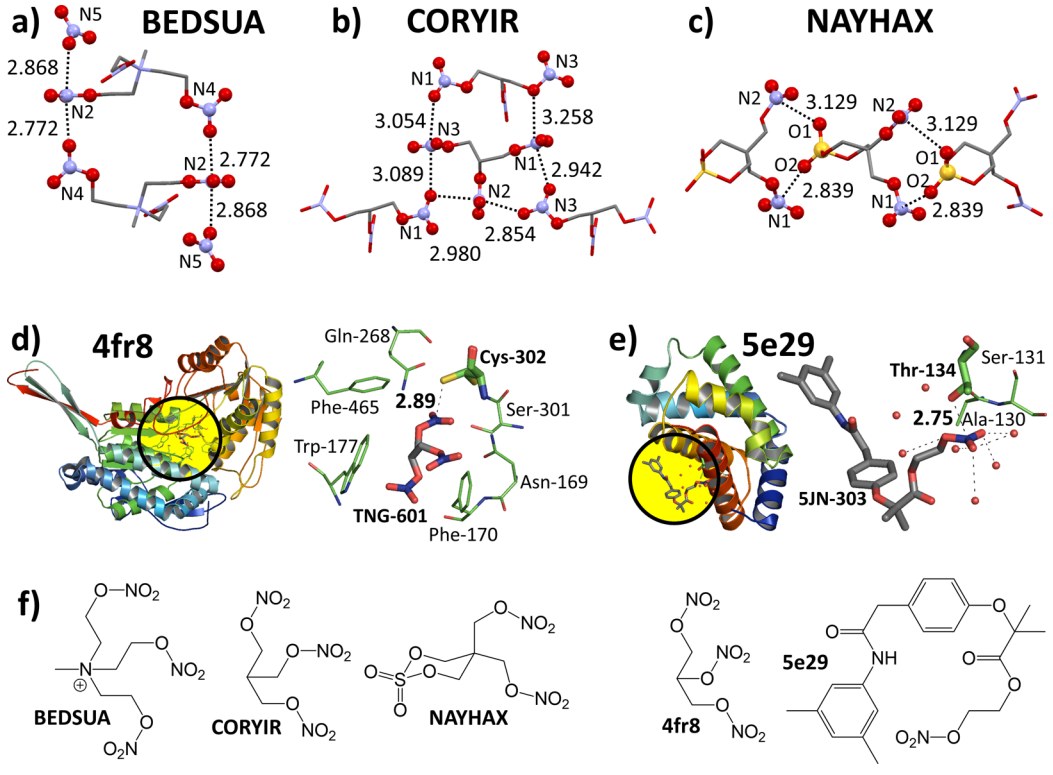


Figure 4. Examples of crystal structures found within the CSD (a–c) and the PDB (d and e) where π-hole interactions seem evident. The chemical structures of relevant nitro ester molecules involved are shown in f. The PDB structures both concern a ligand within chain A of the parent structure and the residues displayed are those within a 4 Å envelope around the relevant CO–NO_2 unit. Color code: carbon = gray or green, hydrogen = white, nitrogen = blue, oxygen = red, sulfur = yellow. Distances are in Å.

acetonitrile and dimethyl ether the anticipated π-hole complexes were always obtained with estimated binding energies ranging from −3.4 kcal/mol in [**10**···$NCCH_3$] to −5.3 kcal/mol in [**6**···$O(CH_3)_2$] (see Figure 2e). This is in the range of moderate hydrogen bonding (4–15 kcal/mol).[27] In general, the binding energies with $(CH_3)_2O$ are larger than those with CH_3CN by about 1 kcal/mol. The "Atoms in Molecules"[26] analysis of some complexes shows the formation of bond paths and bond critical points that connect the electron donor atom and the π-hole bearing atom (mostly N, see Figure S1) thus confirming the existence of π-hole interactions in these systems. Moreover, the Laplacian of the electron density

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is positive in all complexes, which is an indication of the closed shell (noncovalent) nature of the interaction.

The above computations clearly suggest that π-hole interactions with R−NO_2 moieties are energetically viable. It is known that for R = CH_3 and for aromatic nitro-compounds such interactions exist and can be directional in the solid state.[8] We thus set out to evaluate the crystal structures deposited in the CSD[28] for any evidence of π-hole interactions with other nitro-bearing molecules (CSD version 5.37 (November 2015) including 2 updates was queried using the ConQuest user interface. The procedure to determine the XYZ-coordinates of the hits is explained in the Supporting Information). Of the R− groups considered it appeared that there are 175 crystallographic information files (CIFs) containing the well-known nitrate ester functionality (i.e., R = OCX_3)[30,31] which could thus be used for a more detailed evaluation.[29]

Initially, a data set was extracted from the CSD that contained "hits" where the intermolecular distance between the nitrate ester's N atom and any electron rich atom is ≤5 Å. In total, 3347 such hits were found (one CIF can contain multiple hits). Interestingly, of the data directly above/below N (346 hits), about a third (114 hits) has overlapping van der Waals shells (Figure S4). It appeared that the majority of these data (i.e., 97 of 114) consisted of structures involving an sp^2 oxygen atom. Hence the [CO−NO_2···O═X] pair was scrutinized further and, in total, 1700 such hits were found within 154 CIFs. As is evident from Figure 3a and b, sp^2 O atoms seem to cluster near the nitrate ester's N atom, roughly above the area characterized by a parallel displacement of ≤1 Å from N (black circle in Figure 3b). These data were further scrutinized by plotting the percentage of data as a function of the van der Waals corrected N···O distance, as shown in Figure 3c. It appears that a Gaussian-shaped clustering is observed around the van der Waals benchmark and that 50% of these molecular pairs involve overlapping van der Waals shells. When considering hydrogen atoms as the interacting partner (Figure S5) the data was scattered around the oxygen periphery and merely 1.7% of the data was found directly above/below N (compare with 11.5% for sp^2 O, Figure 3b). This suggests that in the solid state nitrate esters can be directional π-hole donors.

The structures with a N···O distance of ≤2.87 Å (i.e., more than 0.2 Å van der Waals overlap) were inspected manually and three of these (BEDSUA,[32] CORYIR,[33] and NAYHAX[34]) are shown in Figure 4a−c. In all three structures, a network of short intermolecular N···O distances is observed that likely determine the crystal packing of these small molecules.

The Brookhaven protein databank (PDB) was also consulted for ligands bearing a nitrate ester. While merely four such structures were found (3k2f,[35] 3ni5,[36] 4fr8,[37] and 5e29[38]), in two of these the nitrate ester's N atom is in close contact with an electron rich atom. In 4fr8 (Figure 4a) the tris-nitrate ligand TNG-601 is buried within an aromatic-rich pocket and one of the nitrate N atoms is in close contact (2.89 Å) with a S atom from the disordered cysteine 302. Within 5e29 (Figure 4e) the nitrate ester portion of ligand 5JN-303 is embedded within a water pocket and a short N···O distance of 2.75 Å is observed with threonine 134. Both these distances are well within the van der Waals benchmark of, respectively, 3.35 Å for N+S and 3.07 Å for N+O.[39]

CONCLUDING REMARKS

The computations, CSD analyses, and crystal structures highlighted in this work support the idea that π-hole interactions with nitro-compounds can be a directional force in the solid state. In particular it was found that a nitrate ester can be a potent and directional π-hole. This implies that the interaction might be rationally exploited in crystal engineering, much like hydrogen and halogen bonding. Our finding may also help rationalize biologically relevant binding and transport phenomena involving nitrate esters and aid in drug design generally. Indeed, not only are organic nitrate esters natural molecules,[40,41] but they have also been reported as inhibitors,[42−45] and are widely used as vasodilators to treat cardiovascular conditions.[46]

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.cgd.6b00989.

Details of computations and CSD analysis (PDF)

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Notes

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